

Laboratory Work 5 Activity | Applying DAX Fundamentals in Power BI: From Measures to Contextual Analysis

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IS 107 – Business Intelligence

DATA USED.



LW5_Cookie Types.xlsx

Authors: Kevin Stratvert



LW5_Customers.xlsx

Authors: Kevin Stratvert



LW5_Orders.xlsx

Authors: Kevin Stratvert

Measure formulas.

Part 1:

Formula

```
1 Total Number of Units Sold = SUM(Orders[Units Sold])
```

Result:

Total Number of Units Sold
1,125,806.00

Name	Total Number of Units Sold
ABC Groceries	221,023.00
ACME Bites	329,676.50
Park & Shop Convenience Stores	179,447.50
Tres Delicious	128,769.00
Wholesome Foods	266,890.00
Total	1,125,806.00

Measure formulas.

Part 2:

Formula

```
1 Count of Orders = COUNTROWS(Orders)
```

Result:

Name	Count of Orders
ABC Groceries	132
ACME Bites	206
Park & Shop Convenience Stores	114
Tres Delicious	92
Wholesome Foods	156
Total	700

Part 3:

Formula

```
1 Distinct Customers = DISTINCTCOUNT(Orders[Customer ID])
```

Result:

Distinct Customers
5

Measure formulas.

Part 4:

Formula

```
1 Total Profit = SUM(Orders[Revenue]) - SUM(Orders[Cost])
```

Result:

Name	Distinct Customers	Total Profit
ABC Groceries	1	523,317
ACME Bites	1	828,379
Park & Shop Convenience Stores	1	423,499
Tres Delicious	1	302,199
Wholesome Foods	1	639,677
Total	5	2,717,071

Part 5:

Formula

```
1 Profit = Orders[Revenue] - Orders[Cost]
```

Result:

Customer ID	Order ID	Product	Units Sold	Date	Revenue	Cost	Profit
3	266868	Chocolate Chip	292	Saturday, February 1, 2020	1460	584	876
3	140794	Chocolate Chip	974	Saturday, February 1, 2020	4870	1948	2922
3	684759	Chocolate Chip	2518	Monday, June 1, 2020	12590	5036	7554
3	183251	Chocolate Chip	1513	Tuesday, December 1, 2020	7565	3026	4539
3	600124	Chocolate Chip	1030	Friday, May 1, 2020	5150	2060	3090
3	562219	Chocolate Chip	1514	Saturday, February 1, 2020	7570	3028	4542
3	387444	Chocolate Chip	787	Monday, June 1, 2020	3935	1574	2361
3	365463	Chocolate Chip	1728	Friday, May 1, 2020	8640	3456	5184
3	505159	Chocolate Chip	2145	Tuesday, October 1, 2019	10725	4290	6435
3	300303	Chocolate Chip	1084	Tuesday, December 1, 2020	5420	2168	3252
3	578401	Chocolate Chip	689	Monday, June 1, 2020	3445	1378	2067
3	365552	Chocolate Chip	1570	Monday, June 1, 2020	7850	3140	4710
3	239419	Chocolate Chip	4251	Wednesday, January 1, 2020	21255	8502	12753
3	549329	Chocolate Chip	2918	Friday, May 1, 2020	14590	5836	8754
3	288851	Chocolate Chip	2988	Wednesday, July 1, 2020	14940	5976	8964
3	255145	Chocolate Chip	2074	Tuesday, September 1, 2020	10370	4148	6222
3	538134	Chocolate Chip	1514	Tuesday, October 1, 2019	7570	3028	4542
3	817134	Chocolate Chip	274	Tuesday, December 1, 2020	1370	548	822
3	697568	Chocolate Chip	1138	Tuesday, December 1, 2020	5690	2276	3414
3	508782	Chocolate Chip	2002	Sunday, March 1, 2020	14060	5084	8976

Measure formulas.

Part 6:

Formula

```
1 Profit MArgin % = [Total Profit] / sum(Orders[Revenue])
```

Result:

Cookie Type	Profit MArgin %
Chocolate Chip	60.00%
Fortune Cookie	50.00%
Oatmeal Raisin	56.00%
Snickerdoodle	62.50%
Sugar	58.33%
White Chocolate Macadamia Nut	54.17%
Total	57.93%

Part 7:

Formula

Result:

Measure formulas.

Part 7:

Formula

```
1 Total Profit 2 = SUMX('Cookie Types', 'Cookie Types'[Units Sold] * ('Cookie Types'[Revenue Per Cookie] - 'Cookie Types'[Cost Per Cookie]))
```

Result:

Cookie Type	Total Profit 2
White Chocolate Macadamia Nut	527,879.63
Sugar	295,370.25
Snickerdoodle	367,115.00
Oatmeal Raisin	434,882.00
Fortune Cookie	77,099.00
Chocolate Chip	1,014,718.50
Total	2,717,064.38

Measure formulas.

Part 8:

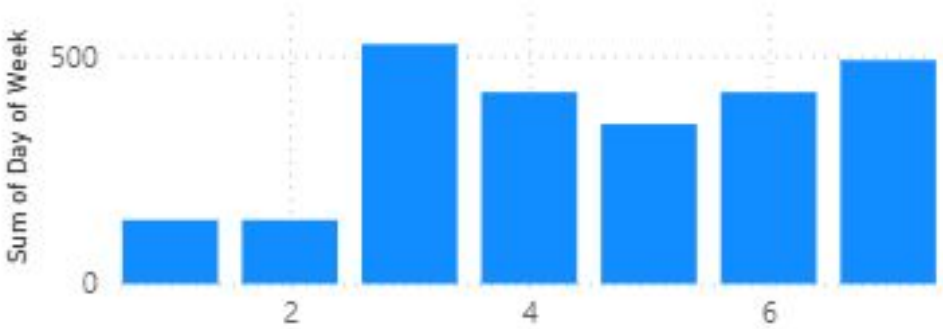
Formula

```
1 Day of Week = WEEKDAY(Orders[Date],1)
```

Result:

Customer ID ▾	Order ID ▾	Product ▾	Units Sold ▾	Date ▾	Revenue ▾	Cost ▾	Profit ▾	Day of Week ▾
3	266868	Chocolate Chip	292	Saturday, February 1, 2020	1460	584	876	7
3	140794	Chocolate Chip	974	Saturday, February 1, 2020	4870	1948	2922	7

Sum of Day of Week by Day of Week



Measure formulas.

Part 9:

Formula

```
1 Has Chocolate = FIND("Chocolate", Orders[Product], 1, 0)
```

Result:

Customer ID	Order ID	Product	Units Sold	Date	Revenue	Cost	Profit	Day of Week
3	266868	Chocolate Chip	292	Saturday, February 1, 2020	1460	584	876	7
3	140794	Chocolate Chip	974	Saturday, February 1, 2020	4870	1948	2922	7
3	684759	Chocolate Chip	2518	Monday, June 1, 2020	12590	5036	7554	2
3	182251	Chocolate Chip	1512	Tuesday, December 1, 2020	7565	2926	4639	3

Measure formulas.

Part 10:

Formula

```
1 Has Chocolate = IF(FIND("Chocolate", Orders[Product], 1, 0)>0 ,"Has Chocolate", "No Chocolate")
```

Result:

Customer ID	Order ID	Product	Units Sold	Date	Revenue	Cost	Profit	Day of Week	Has Chocolate
3	266868	Chocolate Chip	292	Saturday, February 1, 2020	1460	584	876	7	Has Chocolate
3	140794	Chocolate Chip	974	Saturday, February 1, 2020	4870	1948	2922	7	Has Chocolate

Measure formulas.

Part 11:

Formula

```
1 Chocolate Chips with over 500 units = CALCULATE(COUNTROWS(Orders), Orders[Units Sold]>500,Orders[Product] == "Chocolate Chip")
```

Result:

